Faculty of Computing and Information Technology (FCIT)

Database Systems - ASSIGNMENT#2

BS(SE) - Fall 2023 (Semester Spring 2025)

Total Marks: 100 Deadline: **May 20, 2025**

Instructions

- You must complete all tasks individually. Absolutely **NO** collaboration is allowed.
- Any traces of plagiarism/cheating would result in an "F" grade in this course.
- Late submissions will **NOT** be accepted, in any case.
- You are also required to submit pdf file to the respective TAs.
- Name of your file should be **FD_Norm_RollNumber** e.g. FD_Norm_BSEF24M0XX. Use the same title as a subject of your email for submission.

Books

A. Thomas Connolly, "Database Systems: A Practical Approach to Design, Implementation and Management (6th Ed.)" 14.19, 14.20
B. Jeffrey Hoffer, "Modern Database Management" Design, Implementation, Management, 12th Edition" 4.57, 4.58

Section-I

Quesition#1[15]

a) The set of functional dependencies for a relation R(A,B,C,D,E,F,G) is:

$$F = \{AD \rightarrow BF,\ CD \rightarrow EGC,\ BD \rightarrow F,\ E \rightarrow D,\ F \rightarrow C,\ D \rightarrow F\}$$

i. Find all candidate keys.
ii. Find F-closure.

b) The set of functional dependencies for a relation R(A,B,C,D,E,F,G,H,I,J) is:

$$F = \{AB \rightarrow C,\ A \rightarrow DE,\ B \rightarrow F,\ F \rightarrow GH,\ D \rightarrow IJ\}$$

i. What is key for R?
ii. Decompose R into 2NF and then into 3NF.

Quesition#2[10]

The set of functional dependencies for relation R(A,B,C,D,E) is:

$$F = \{AB \rightarrow C,\ AB \rightarrow D,\ D \rightarrow A,\ BC \rightarrow D,\ BC \rightarrow E\}$$

a. Find all candidate keys.
b. Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF).
c. Is this relation in BCNF? If not, show all dependencies that violate it.
d. Is this relation in 3NF? If not, show all dependencies that violate it.

Section-II

Question#3 [10]

Book A: Problem 14.14

Quesition#4 [10]

A leasing company lease flats to its clients. The sample data of the company is given in the table below. A place number (placeNo) uniquely identifies each single room in all flats and is used when leasing a room to a student. Perform the following tasks:

- Identify the functional dependencies that exist between the columns of the table and identify the primary key and any alternate keys (if present) for the table.
- Describe why the table is not in 3NF.
- The table is susceptible to update anomalies. Provide examples of how insertion, deletion, and modification anomalies could occur on this table.

leaseNo	bannerID	placeNo	fName	lName	startDate	finishDate	flatNo	flatAddress
10003	B017706	78	Jane	Watt	01/09/2010	30/06/2011	F56	34 High Street, Paisley
10259	B017706	88	Jane	Watt	01/09/2011	30/06/2012	F78	111 Storrie Road, Paisley
10364	B013399	89	Tom	Jones	01/09/2011	30/06/2012	F78	111 Storrie Road, Paisley
10566	B012124	102	Karen	Black	01/09/2011	30/06/2012	F79	120 Lady Lane, Paisley
11067	B034511	88	Steven	Smith	01/09/2012	30/06/2013	F78	111 Storrie Road, Paisley
11169	B013399	78	Tom	Jones	01/09/2012	30/06/2013	F56	34 High Street, Paisley

Quesition#5 [10]

Book B: Problem 4.54

Quesition#6 [15]

Book B: Problem 4.58 (parts a,b,c)

Question# 7[15]:

The University Enrollment System tracks information about students, courses, instructors, departments, and enrollment details in a single table. This table is poorly designed and contains redundant and repeated data. Each time a student enrolls in a course, all related information about the student, instructor, course, and department is repeated in a new row. This leads to issues such as redundancy, anomalies (update, insertion, and deletion), and a lack of scalability.
Normalize the table given below by organizing the data into multiple related tables to eliminate redundancy, maintain data integrity, and scale efficiently for thousands of students, courses, and departments.

Attributes:
RollNo , Name, Email, Degree, CourseCode (CC) , CourseTitle (CT), InstructorID (IID), InstructorName (IN), Designation, Deptno (DN), DName, DHead, EnrollmentDate (EDate), Grade, ClassRoom (CR), Schedule

Sample Data

Roll No	Name	Degree	CC	Course Title	IID	Instructor Name	Instructor Email	DN	DName	DHead	EDate	Grade	CR	Schedule
1	Alice Smith	BS(SE)	101	Data Structures	501	Dr. John	john@mail.com	201	Software Engineering	Dr. Brown	01/15/2025	A	101	Mon/Wed 9:00-10:30
2	Bob Johnson	BS(CS)	102	Operating Systems	502	Dr. Susan	susan@mail.com	202	Computer Science	Dr. White	01/16/2025	B+	102	Tue/Thu 11:00-12:30
3	Nick Knight	BS(CS)	102	Operating Systems	502	Dr. Susan	susan@mail.com	202	Computer Science	Dr. White	01/16/2025	B	102	Tue/Thu 11:00-12:30

Quesition#8 [15]

VisionEstate is a property management company overseeing various residential properties across different cities in Pakistan. The company conducts routine property inspections to ensure the upkeep of these residences. When staff members need to carry out these assessments, they are assigned a corporate vehicle for use during the inspection day. However, a single vehicle may be designated to multiple staff members as needed throughout the workday. A staff member might assess numerous properties on a specific date, yet each property undergoes assessment only once on that particular day. Samples of the VisionEstate Property Evaluation Report are depicted in table below. IDate and ITime stands for InsprectionDate and InsprectionTime respectively.

Sample Data

PropertyNo	PAddress	IDate	ITime	Comments	StaffNo	SName	CarReg
PK12	10-A, Model Town, Lahore	05-07-22	09:30	Roof repair needed	SG24	Saim Ghani	LM-5678
		15-08-22	11:00	In good order	AJ11	Ali Javed	PB-9876
		1-10-22	9:00	Bathroom tiles replacement	SG24	Saim Ghani	LW-3421
PK17	8-B, Gulberg, Karachi	02-06-22	10:45	Replace living room carpet	FA09	Farah Ahmed	LM-5678
		25-08-22	12:30	Good condition	SG24	Saim Ghani	PB-9876

a) Find all candidate keys.
b) Identify the best normal form that the above table satisfies and provide reasoning (1NF, 2NF, 3NF, or BCNF).
c) Is this relation in BCNF? If not, show all dependencies that violate it.
d) Is this relation in 3NF? If not, show all dependencies that violate it.
e) Discuss potential issues related to updates on this table, including insertion, deletion, and modification anomalies.
f) Describe and illustrate the process of normalizing the relation shown in above Table to 4NF.